

PHYS 143 – Mechanics and Thermal Physics

Motion

*How do we describe where an object is, how it moves,
and how its motion changes?*

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By the end of this week, you should be able to:

1. Distinguish position, displacement, velocity and acceleration, and represent them with diagrams, graphs and equations.
2. Calculate velocity and acceleration by differentiation, and recover them by integration with initial conditions.
3. Select a kinematic model, and state when it is valid.
4. Solve free-fall, projectile, circular and relative-motion problems.

The observable skills are in the checklist at the end of the lecture. **Reading:** Young and Freedman, Ch. 2, §2.1–2.6 and Ch. 3, §3.1–3.5.

Three contacts:

1 DESCRIBE

How do we represent motion?

2 MODEL

How do we describe changing motion mathematically?

3 APPLY

How do the same ideas describe more complex motion?

Practice: concept check → lecturer example
→ independent problem → application problem

What to master, and what supports it

Core skills you should master

- distinguish the six quantities of motion;
- choose an origin and positive direction;
- differentiate $\vec{r}(t)$ to find $\vec{v}$ and $\vec{a}$;
- integrate $a(t)$ with initial conditions;
- read and sketch $x-t$, $v-t$, $a-t$ graphs;
- decide from the signs of v_x , a_x if speed rises;
- choose a constant- a equation, say when it applies;
- solve free fall, choosing the physical root;
- solve projectiles with components linked by time;
- explain and calculate radial acceleration;
- add velocities with the subscript chain.

Supporting tools

- the parametric form of a trajectory;
- tangential and normal acceleration on a general curve;
- derivations and the calculus refresher (supplement).

Supporting slides carry the label *supporting tool* in their titles. They stay in the deck for reference; the core skills are what the problems test first.

Not marked. Answer individually, then we review.

1. Give the signs of the x - and y -components of a 20 m/s vector at 30° above the $-x$ axis.
2. Differentiate $x = 3t^2 - 4t + 2$ and evaluate the result at $t = 2 \text{ s}$.
3. Integrate $a = 6t$, given $v = 0$ at $t = 0$.
4. Solve $4.9t^2 - 12t - 3 = 0$. Which root can be a time?
5. What do the slope and the area under a straight-line graph segment mean?
6. Convert 72 km/h to m/s .

These are the Week 1 and calculus skills this week depends on. A calculus refresher is in the supplementary notes.

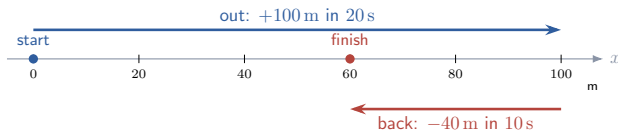
1. 150° from $+x$: x negative, y positive
(-17.3 , $+10.0$).
2. $dx/dt = 6t - 4$, so 8 m/s at $t = 2 \text{ s}$.
3. $v = 3t^2$.
4. $t = \frac{12 \pm \sqrt{144 + 58.8}}{9.8} = 2.68 \text{ s}$ or -0.23 s . Only the positive root can be a time after $t = 0$.
5. Slope = rise/run; area = base $\times$ height (triangle: half of it).
6. $72/3.6 = 20 \text{ m/s}$.

Contact 1: Describe

How do we represent motion?

Opening problem

A runner goes 100 m east in 20 s, turns, and runs 40 m back west in 10 s. Where is the runner? Over the whole journey, how has the position changed, and how rapidly and in what direction?



Predict first

Before any calculation: is the average velocity positive or negative? Is it larger or smaller than the average speed?

But what if... “how far did the runner go?” has two honest answers, 60 m from the start or 140 m of road? We need words that tell them apart: the **vocabulary of motion**.

Defining the vocabulary of motion (1): where, and how far

Quantity	What it is	What it measures	SI unit	Representation
Position	location relative to a chosen origin	where the object is	m	a point on an axis; the vector $\vec{r}$
Displacement	final position minus initial position (a vector)	how far and which way the position has changed	m	arrow from start to finish
Distance	total length of the path travelled (a scalar)	how much ground was covered	m	length along the path

Interpret: position answers *where is it?*; displacement answers *how has the position changed?*; distance answers *how much ground was covered?*

But what if... the journey also asks *how rapidly* the position changes? Words are needed for rates of change: **velocity, speed and acceleration.**

Defining the vocabulary of motion (2): how fast, and how it changes

Quantity	What it is	What it measures	SI unit	Representation
Velocity	rate of change of position (a vector)	how fast, and in which direction, the position changes	m/s	tangent to the path; slope of $x-t$
Speed	the magnitude of the velocity (a scalar)	how fast, with no direction	m/s	length of the velocity arrow
Acceleration	rate of change of velocity (a vector)	how quickly the speed or the direction changes	m/s ²	arrow; slope of $v-t$

Interpret: velocity answers *how fast, and which way?*; acceleration answers *how is the velocity changing?* Each is defined precisely, with its equation, in the slides that follow.

But what if... the same word names a vector in one place and a number in another? We must fix the **notation** before using these words in equations.

Notation: vector, magnitude and component

Notation used in every lecture

Vector: $\vec{v}$ (bold, with an arrow). **Magnitude:** $v = |\vec{v}| \geq 0$, a number with no direction. For velocity, v is the **speed**.

Scalar component: v_x , a *signed number*: the part of $\vec{v}$ along the x axis you chose.

Component vector: $v_x \hat{i}$. The number v_x is not a vector; $v_x \hat{i}$ is.

The runner, going west

With $+x$ east:

$v_x = -4.0 \text{ m/s}$ (a signed number)

$\vec{v} = -4.0 \hat{i} \text{ m/s}$ (a vector)

$v = |\vec{v}| = 4.0 \text{ m/s}$ (the speed)

Careful

v_x can be negative; the speed v cannot. In 1D we write x , v_x , a_x for signed components along the chosen axis.

But what if... the sign of v_x depends on which way we call positive? Then position and displacement need an **origin** and an **axis** chosen first.

Position and displacement in one dimension

- **Position** $x(t)$: where the object is, relative to a chosen *origin* along a chosen *axis*.

- **Displacement:**

$$\Delta x = x_2 - x_1 \quad (\text{final} - \text{initial})$$

It can be positive, negative or zero. It does not depend on the path.

- **Distance travelled:** the total path length, a scalar greater than or equal to zero.

Physical meaning

Position depends on the origin and axis you choose. Displacement is independent of the origin, provided the same coordinate system is used for both positions. Its sign depends on which direction you call positive.

Link to Week 1

In 1D, Δx is the x -component of $\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$, vector subtraction from last week.

But what if... the runner turns back? x goes $0 \rightarrow 100 \rightarrow 60$ m: $\Delta x = +60$ m, distance 140 m. **Test** the two ideas.

Concept check: distance or displacement?

Which of these can be negative? Which is path-independent?

1. The length of road travelled from Accra to Kumasi.
2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Concept check: distance or displacement?

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1. The length of road travelled from Accra to Kumasi.
2. The change in position from Accra to Kumasi, with x measured north from Accra.
3. The change in position of a lap runner after one complete lap.
4. The odometer reading minus the starting reading.

Answers: (1) distance, never negative. (2) displacement, positive here. (3) displacement = 0, distance = one lap. (4) distance. Displacement depends only on the end points; distance depends on the whole path.

But what if... we measure x westward instead? Every sign in the runner's journey flips ($\Delta x = -60$ m), yet nothing physical changed. We need a habit that fixes the signs first: **Toolkit 1**.

TOOLKIT 1 – Signs and coordinates

WHEN TO USE: any problem with a position, displacement, velocity or acceleration along a line; every later toolkit this week starts here.

PROCEDURE: **Choose** an origin and a positive direction *before* assigning any sign, and mark them on the sketch. **Assign** signs relative to *your* axis: $v_x > 0$ means moving toward $+x$, $v_x < 0$ toward $-x$; the speed is $|v_x|$. A change in anything is final minus initial.

WATCH OUT: reversing the axis flips every sign; the physics does not change.

CHECK: does each sign agree with the sketch and with the motion described?

But what if... we now ask how rapidly the runner's position changed over the 30 s? That needs one number for the whole journey: **average velocity**.

Definitions

$$v_{\text{av},x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

$$\text{average speed} = \frac{\text{distance travelled}}{\Delta t}$$

The signed $v_{\text{av},x}$ can be negative; the average speed cannot.

Back to the runner

Find the average velocity and the average speed for the whole 30 s.

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The signed $v_{\text{av},x}$ can be negative; the average speed cannot.

Back to the runner

Find the average velocity and the average speed for the whole 30 s.

$$\Delta x = +100 - 40 = +60 \text{ m},$$

$$v_{\text{av},x} = +2.0 \text{ m/s}.$$

Distance = 140 m, so average speed = 4.7 m/s.

Evaluate: positive, and smaller than the average speed. The prediction holds.

But what if... the runner never actually moved at +2.0 m/s? The average hides what happened (first +5.0, then -4.0 m/s). Try another case first, then ask about *one instant*.

Question

A dragster is at $x_1 = 19 \text{ m}$ at $t_1 = 1.0 \text{ s}$ and at $x_2 = 277 \text{ m}$ at $t_2 = 4.0 \text{ s}$. Find its average x -velocity.

Predict first

Positive, and roughly what size? (It covers about 250 m in about 3 s.)

Question

A dragster is at $x_1 = 19 \text{ m}$ at $t_1 = 1.0 \text{ s}$ and at $x_2 = 277 \text{ m}$ at $t_2 = 4.0 \text{ s}$. Find its average x -velocity.

Predict first

Positive, and roughly what size? (It covers about 250 m in about 3 s.)

Identify: target $v_{\text{av},x}$; known two positions and times. **Set up:** x along the track, positive forward.

Execute:
$$v_{\text{av},x} = \frac{277 - 19}{4.0 - 1.0} \text{ m/s} = \frac{258}{3.0} = 86 \text{ m/s}.$$

Evaluate: positive, units m/s, consistent with the estimate ($\approx 83 \text{ m/s}$). ✓

Cf. Young and Freedman, Example 2.1.

But what if... the dragster was slower than 86 m/s at the start and faster at the end? What was its velocity at exactly $t = 2.0 \text{ s}$? → **instantaneous velocity**.

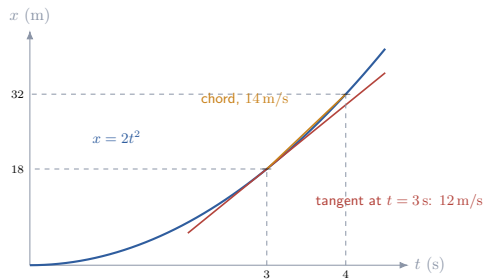
From average to instantaneous velocity

A particle moves with $x = 2t^2$ (SI units).

Shrink the interval after $t = 3$ s:

Δt (s)	$v_{\text{av},x}$ (m/s)
1.0	14.0
0.5	13.0
0.1	12.2
0.01	12.02

The chord slope approaches the tangent slope:
12 m/s.



But what if... the table only *approaches* 12 m/s and we had to guess the limit? Make the limit exact for any t : the **derivative**.

Rule: instantaneous velocity

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad \text{the slope of the tangent to the } x-t \text{ graph}$$

Reading the graph:

- positive slope $\rightarrow v_x > 0$ (moving in $+x$)
- negative slope $\rightarrow v_x < 0$
- zero slope $\rightarrow$ momentarily at rest
- steeper $\rightarrow$ greater speed $|v_x|$

Two different graphs

An $x-t$ graph is *not* the path of the object. The height of the graph is the position, not the speed.

But what if... we want to know exactly what each symbol, sign and unit in these two velocity equations means? $\rightarrow$ **equation anatomy**.

EQUATION – average velocity

$$v_{\text{av},x} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1}$$

Meaning. the net change of position per unit time over an interval.

Variables. Δx is final minus initial position; Δt is the elapsed time.

Units. m/s.

Interpretation. the sign gives the direction of the net motion; it is not the average speed.

Example. the dragster, +86 m/s; the runner, +2.0 m/s.

EQUATION – instantaneous velocity

$$v_x = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Meaning. the velocity at one instant: the limit of the average over ever shorter intervals.

Variables. $x(t)$ is the position as a function of time.

Units. m/s.

Interpretation. the slope of the tangent to the x - t graph; its sign is the direction, its size is the speed.

Example. $x = 2t^2$: $v_x = 4t = 12 \text{ m/s}$ at $t = 3 \text{ s}$, as the table of chords suggested.

Concept check: velocity before any algebra

Decide, and sketch an $x-t$ graph where it helps

1. Can the average speed over a journey be smaller than $|v_{av,x}|$?
2. Can the velocity be zero while the object is not at the origin?
3. Can the velocity be zero at one instant and the object still be moving a moment later?
4. Can the speed be constant while the velocity changes?

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Answers

1. **No.** Distance $\geq |\Delta x|$, so average speed $\geq |v_{\text{av},x}|$.
2. **Yes.** Velocity says how position is changing, not where it is.
3. **Yes.** A ball at the top of its flight has $v = 0$ only for an instant.
4. **Yes.** Constant speed on a curve: the direction of $\vec{v}$ changes (Contact 3).

But what if... the motion is not along one line, such as a ball thrown across a field? Position then needs x , y (and z): a **vector**.

From a number line to vectors

In two or three dimensions the position becomes a vector:

$$\vec{r}(t) = x(t) \hat{i} + y(t) \hat{j} + z(t) \hat{k}$$

Rule: displacement and velocity as vectors

$$\Delta \vec{r} = \vec{r}_2 - \vec{r}_1 \quad \vec{v}_{\text{av}} = \frac{\Delta \vec{r}}{\Delta t}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} + \frac{dz}{dt} \hat{k}$$

Physical meaning

The velocity vector is **tangent to the path** and points in the instantaneous direction of motion. $\vec{v}_{\text{av}}$ has the direction of $\Delta \vec{r}$.

Week 1 tools

Components, signs, magnitude

$v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ and quadrants carry over.

But what if... $\vec{r}(t)$ is given by formulas for $x(t)$, $y(t)$, $z(t)$? Differentiate each component: **Toolkit 2**.

✂ TOOLKIT 2 – The calculus of motion

WHEN TO USE: the position (or velocity) is given as formulas in t and you need the velocity (or acceleration). The reverse step, integration, is added in Contact 2.

$$\text{position } \vec{r}(t) \xrightarrow{d/dt} \text{velocity } \vec{v}(t) \xrightarrow{d/dt} \text{acceleration } \vec{a}(t)$$

PROCEDURE: Differentiate component by component: $\frac{d}{dt}t^n = nt^{n-1}$,
 $\frac{d}{dt} \cos \omega t = -\omega \sin \omega t$, $\frac{d}{dt} \sin \omega t = \omega \cos \omega t$. **Substitute** the time *after* differentiating.

WATCH OUT: the argument of sin and cos is in radians; ωt is dimensionless.

CHECK: every derivative divides by seconds, so the units must change accordingly; the direction of $\vec{v}$ matches the sketch of the path.

Question

A particle has position (SI units, t in seconds)

$$\vec{r} = 2.0 \cos(\pi t) \hat{i} + 2.0 \sin(\pi t) \hat{j} + 1.0 t^2 \hat{k}$$

Find its velocity at $t = 0.5$ s.

Predict first

At $t = 0$ the particle is at $(2.0, 0, 0)$. Sketch the motion. In which direction is it moving at $t = 0.5$ s, and is z increasing?

Identify: target $\vec{v}(0.5\text{ s})$; known $\vec{r}(t)$. **Set up:** differentiate each component (Toolkit 2).

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$$\vec{v} = \frac{d\vec{r}}{dt} = -2.0\pi \sin(\pi t) \hat{i} + 2.0\pi \cos(\pi t) \hat{j} + 2.0t \hat{k}$$

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Execute at $t = 0.5\text{ s}$: $\sin(\pi/2) = 1$, $\cos(\pi/2) = 0$, so

$$\vec{v} = (-6.28 \hat{i} + 0 \hat{j} + 1.0 \hat{k}) \text{ m/s}, \quad v = \sqrt{6.28^2 + 1.0^2} = 6.36 \text{ m/s}$$

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Evaluate: at $t = 0.5\text{ s}$ the particle is at $(0, 2.0, 0.25)\text{ m}$, the top of the circle; moving in $-x$ is counter-clockwise from above, and z increases. This agrees with the prediction. Units m/s. ✓

The learning point is differentiating component by component; do not dwell on the geometry. The path is a helix: a circle in the xy -plane combined with steadily increasing z (cf. Young and Freedman, §3.1).

Independent problem

A particle has $\vec{r} = (3.0t^2) \hat{i} + (4.0t) \hat{j}$ (SI units). Find its velocity at $t = 2.0$ s, its speed, and the direction of motion.

Predict first

Sketch first. Is the particle moving mainly in x or y at $t = 2$ s? Speeding up or slowing down in x ?

Your turn: velocity from position

Independent problem

A particle has $\vec{r} = (3.0t^2) \hat{i} + (4.0t) \hat{j}$ (SI units). Find its velocity at $t = 2.0$ s, its speed, and the direction of motion.

Predict first

Sketch first. Is the particle moving mainly in x or y at $t = 2$ s? Speeding up or slowing down in x ?

$$\vec{v} = 6.0t \hat{i} + 4.0 \hat{j} \Rightarrow \vec{v}(2) = (12 \hat{i} + 4.0 \hat{j}) \text{ m/s. } v = \sqrt{144 + 16} = 12.6 \text{ m/s.}$$
$$\theta = \tan^{-1}(4.0/12) = 18.4^\circ \text{ above } +x \text{ (quadrant I).}$$

Evaluate: mainly x , as predicted; units m/s. ✓

But what if... we are given only a *graph* of $x(t)$, with no formula to differentiate? Then slopes must be read off the picture: **Toolkit 3**.

Concept check: reading an $x-t$ graph

Question

Of the four points P, Q, R and S:

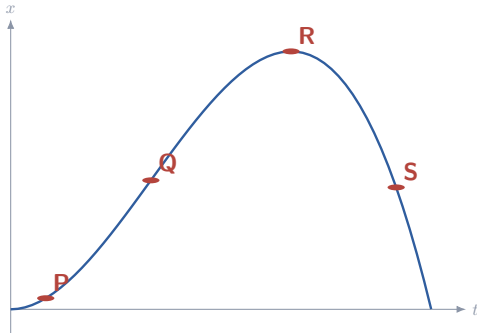
(a) at which is v_x **most positive**?

(b) at which is the **speed** greatest?

Predict first

Decide before the next slide. Which point has the steepest positive slope?

Which the steepest slope of either sign?

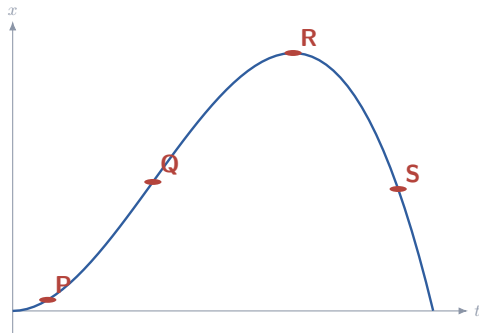


Concept check: solution

This curve is $x = 2t^2 - t^3/3$, so $v_x = 4t - t^2$:

	t (s)	slope	v_x (m/s)
P	0.5	up	+1.75
Q	2	steepest up	+4.0
R	4	flat	0
S	5.5	steepest down	-8.25

(a) Q, (b) S: the speed is $|v_x|$, so a steep *negative* slope counts.



TOOLKIT 3 – Reading motion graphs

WHEN TO USE: the motion is given, or wanted, as a graph rather than a formula.

Graph	slope gives	area gives
$x-t$	velocity v_x	(not used)
v_x-t	acceleration a_x	displacement Δx (<i>Contact 2</i>)
a_x-t	rate of change of a_x	change in velocity Δv_x (<i>Contact 2</i>)

PROCEDURE: Translate both ways: graph $\rightarrow$ description of the motion, and description $\rightarrow$ sketched graph. Read a slope as the vertical change divided by the horizontal change, with units.

WATCH OUT: the graph is not the path; an upward graph does not mean the object moves upward.

CHECK: does each segment's sign and steepness agree with your motion description?

Common mistakes and the correct habit

Mistake

Reading an $x-t$ graph as the path of the object.

Confusing distance with displacement, speed with velocity.

Writing v when a signed component v_x is meant.

Assigning signs before choosing the axes.

Using the average velocity as the instantaneous velocity.

Subtracting initial from final the wrong way round.

Instead

The height is the position; the slope is the velocity.

Displacement and velocity are signed; distance and speed are not.

$v = |\vec{v}| \geq 0$ is the speed; v_x may be negative.

Choose the origin and the positive direction first, and mark them.

Instantaneous velocity is the slope of the tangent, not of the chord.

A change is always final minus initial.

Contact 1: what we can now answer

Where is it? → position $\vec{r}(t)$

How has its position changed? → displacement $\Delta\vec{r}$

How fast and in what direction is its position changing? → velocity $\vec{v} = d\vec{r}/dt$

Connect: Lectures 3 and 4

The slope and area of motion graphs return in Lecture 4, where the area under a force–position graph is the work done. Lecture 3 relates the acceleration you meet next to the force on the object.

But what if... the runner *turns round*? The velocity changes from $+5.0$ to -4.0 m/s. Suppose the turn takes 2.0 s: how rapidly is the velocity changing, and what does the sign of that rate mean? → **Contact 2: acceleration.**

Contact 2: Model

How do we describe changing motion mathematically?

From describing motion to modelling how it changes

So far we have described where an object is and how its position changes. We now ask a new question: **how is the velocity itself changing, and can we use that to predict the motion?**

A new question

From describing motion to modelling how it changes

So far we have described where an object is and how its position changes. We now ask a new question: **how is the velocity itself changing, and can we use that to predict the motion?**

How fast does v change?

acceleration, and the sign rule for speeding up

Given $a(t)$, find v and x

integration with initial conditions

Constant a

constant-acceleration equations and free fall

Poll

Can an object have a **negative** velocity and still be **speeding up**?

Predict first

A car moves in the $-x$ direction and the driver presses the accelerator. Is its velocity negative? Is it speeding up? Is its acceleration positive or negative?

Poll

Can an object have a **negative** velocity and still be **speeding up**?

Predict first

A car moves in the $-x$ direction and the driver presses the accelerator. Is its velocity negative? Is it speeding up? Is its acceleration positive or negative?

Yes. Negative velocity only says the motion is toward $-x$. Whether the object speeds up depends on how v_x and a_x are related; we make this precise in a few slides.

But what if... we try to say this without a definition of a_x itself? First we need the rate of change of velocity: **average acceleration**.

Definition

$$\vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} \quad \text{unit: m/s}^2$$

In 1D: $a_{av,x} = (v_{2x} - v_{1x})/\Delta t$.

$\Delta \vec{v} = \vec{v}_2 - \vec{v}_1 = \vec{v}_2 + (-\vec{v}_1)$: the same vector subtraction as in Week 1.

Physical meaning

Acceleration describes how the **velocity** changes: its magnitude, its direction, or both. It is not only about speed.

Back to the runner

The turn takes 2.0 s and v_x goes from +5.0 to -4.0 m/s:

$$a_{av,x} = \frac{-4.0 - 5.0}{2.0} = -4.5 \text{ m/s}^2.$$

But what if... the runner's $a_x = -4.5 \text{ m/s}^2$ is an *average* over 2.0 s? It cannot say what happens at each instant of the turn. → **instantaneous acceleration**.

Rule: instantaneous acceleration

$$\vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d^2 \vec{r}}{dt^2} \quad a_x = \frac{dv_x}{dt} = \frac{d^2 x}{dt^2}$$

- On a v_x-t graph, a_x is the *slope*.
- On an $x-t$ graph, a_x is the *curvature*:
concave up $\rightarrow a_x > 0$; concave down
 $\rightarrow a_x < 0$.

The chain:

$$x(t) \xrightarrow{d/dt} v_x(t) \xrightarrow{d/dt} a_x(t)$$

But what if... the runner has $a_x < 0$ for the whole turn, yet is first *slowing down* (still moving east) and then *speeding up* (moving west)? The sign of a_x alone cannot tell us which. $\rightarrow$ **anatomy, then the sign rule.**

EQUATION – average acceleration

$$a_{av,x} = \frac{\Delta v_x}{\Delta t} = \frac{v_{2x} - v_{1x}}{t_2 - t_1}$$

Meaning. the change of velocity per unit time over an interval.

Variables. Δv_x is final minus initial velocity; Δt the elapsed time.

Units. m/s^2 , that is, (m/s) per second.

Interpretation. compare its sign with that of v_x : same sign, speeding up; opposite, slowing down.

Example. a car with v_x from +10 to +30 m/s in 4.0 s: $+5.0 \text{ m/s}^2$.

EQUATION – instantaneous

$$a_x = \frac{dv_x}{dt} = \frac{d^2x}{dt^2}$$

Meaning. the rate of change of velocity at one instant.

Variables. $v_x(t)$ is the velocity function; $x(t)$ the position function.

Units. m/s^2 .

Interpretation. the slope of the v_x - t graph; the curvature of the x - t graph.

Example. $x = 2t^2$: $v_x = 4t$, $a_x = 4 \text{ m/s}^2$, constant.

Rule: speeding up or slowing down

Rule: when does speed increase along a line?

Same sign of v_x and $a_x \Rightarrow$ speeding up.

Opposite signs $\Rightarrow$ slowing down.

v_x	a_x	motion
+	+	speeding up (moving toward $+x$)
+	-	slowing down
-	-	speeding up (moving toward $-x$)
-	+	slowing down

Common error

A negative acceleration does not by itself mean slowing down, and a negative velocity does not mean slowing down. Compare the two signs.

But what if... we want to see the same rule on a picture, with no numbers at all? Equal time steps make a **motion diagram**.

Concept check: a motion diagram

Question

The dots show the positions of an object at **equal time intervals**, moving from dot 1 to dot 5 with constant acceleration.

At dot 3, what are the signs of v_x and a_x ?



equal time intervals Δt between dots; the object moves from dot 1 to dot 5

Predict first

Which way is it moving? Are the spacings growing or shrinking?

Concept check: solution

- The object moves toward $-x$, so $v_x < 0$ at dot 3.
- The spacing between dots **shrinks**, so the speed decreases: slowing down.
- Slowing down means v_x and a_x have **opposite** signs, so $a_x > 0$.

Answer

$v_x < 0$ and $a_x > 0$. Check: v_x goes from a larger negative value to a smaller negative one, so $\Delta v_x > 0$.

But what if... the object moves on a curve, so $\vec{v}$ changes direction as well as size? Then $\vec{a}$ has a part *along* the path and a part *across* it. $\rightarrow$ **components of $\vec{a}$** .

Acceleration along and across the path

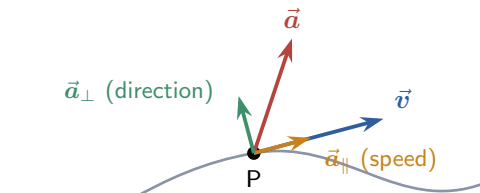
Rule: velocity and the path

$\vec{v}$ is tangent to the path.

$\vec{a}_{\parallel}$ (parallel to $\vec{v}$) changes the **speed**.

$\vec{a}_{\perp}$ (perpendicular to $\vec{v}$) changes the **direction**.

A skier who speeds up on a straight slope has only $a_{\parallel}$. A car taking a bend at constant speed has only $a_{\perp}$ (Contact 3).



But what if... we are handed a v_x-t graph with straight segments instead of a formula? Slopes give a_x , but how do we recover the displacement? → **translating graphs**.

Your turn: translate a $v-t$ graph

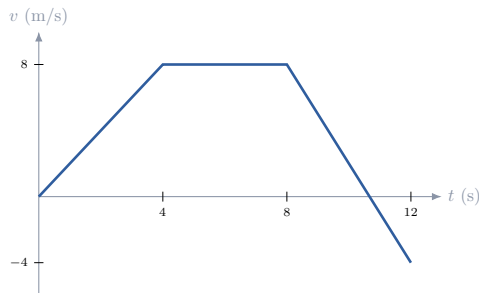
Independent problem

For the motion on the right, find:

- (a) the acceleration in each interval and sketch the a_x-t graph;
- (b) the displacement from 0 to 12 s;
- (c) the distance travelled.

Predict first

Is the displacement larger or smaller than the distance? When does the object reverse?



Solution: slopes and areas

(a) Slopes:

$$0-4\text{ s: } a_x = +8/4 = +2.0\text{ m/s}^2$$

$$4-8\text{ s: } a_x = 0$$

$$8-12\text{ s: } a_x = -12/4 = -3.0\text{ m/s}^2$$

The a_x - t graph is three horizontal steps: +2, 0, -3.

(b), (c) Areas:

$$0-4: \frac{1}{2}(4)(8) = +16\text{ m}$$

$$4-8: 4 \times 8 = +32\text{ m}$$

$$8-10.67: +10.7\text{ m (until } v_x = 0)$$

$$10.67-12: -2.7\text{ m}$$

Result

Displacement = $16 + 32 + 10.7 - 2.7 = +56\text{ m}$; distance = $16 + 32 + 10.7 + 2.7 = 61.3\text{ m}$.

The object reverses at $t = 10.67\text{ s}$, where v_x crosses zero. Area below the axis is negative displacement but positive distance.

TOOLKIT 3 – Reading motion graphs

WHEN TO USE: the motion is given, or wanted, as a graph. Now slope *and* area are used.

Graph	slope gives	area gives
$x-t$	velocity v_x	(not used)
v_x-t	acceleration a_x	displacement Δx
a_x-t	rate of change of a_x	change in velocity Δv_x

PROCEDURE: Areas above the axis add; areas below subtract from the displacement but still add to the distance.

WATCH OUT: where $v_x = 0$ the object may reverse, so split the distance calculation there.

CHECK: do the signs and units of the answer match the graph?

But what if... the graph is a smooth curve $a_x(t)$? The area becomes an **integral**.

TOOLKIT 2 – The calculus of motion (completed)

WHEN TO USE: the acceleration (or velocity) is given as a function of time and you need the velocity and position.

PROCEDURE: **Integrate** from acceleration back to position:

$$v_x(t) = v_{0x} + \int_0^t a_x \, dt' \quad x(t) = x_0 + \int_0^t v_x \, dt'$$

Add the initial conditions v_{0x} and x_0 ; without them the integral is defined only up to a constant. These integrals are the areas under a_x - t (giving Δv_x) and v_x - t (giving Δx), Toolkit 3.

WATCH OUT: do not forget the initial value.

CHECK: $v(0) = v_0$ and $x(0) = x_0$; differentiate your answer to recover the $a(t)$ you started with.

Question

A particle on the x -axis starts at $x_0 = 2.0$ m with $v_{0x} = -3.0$ m/s. Its acceleration is $a_x = (6.0t - 4.0)$ m/s². Find $v_x(t)$, $x(t)$, and the values at $t = 3.0$ s.

Predict first

At $t = 0$, $v_x < 0$ and $a_x < 0$. Is it speeding up or slowing down at first? At $t = 3$ s, what is the sign of a_x ?

Question

A particle on the x -axis starts at $x_0 = 2.0$ m with $v_{0x} = -3.0$ m/s. Its acceleration is $a_x = (6.0t - 4.0)$ m/s². Find $v_x(t)$, $x(t)$, and the values at $t = 3.0$ s.

Predict first

At $t = 0$, $v_x < 0$ and $a_x < 0$. Is it speeding up or slowing down at first? At $t = 3$ s, what is the sign of a_x ?

Execute: $v_x = -3.0 + \int_0^t (6.0t' - 4.0)dt' = -3.0 - 4.0t + 3.0t^2$

$$x = 2.0 + \int_0^t (-3.0 - 4.0t' + 3.0t'^2)dt' = 2.0 - 3.0t - 2.0t^2 + t^3$$

At $t = 3.0$ s: $v_x = -3 - 12 + 27 = +12$ m/s, $x = 2 - 9 - 18 + 27 = 2.0$ m, $a_x = +14$ m/s².

Evaluate: v_x and a_x have the same sign at $t = 0$ (speeding up toward $-x$); v_x then reverses and the particle returns to $x = 2.0$ m. Differentiating x twice gives $6.0t - 4.0$. ✓

Concept check: acceleration before algebra

Decide before any equation

1. Can the velocity be zero while the acceleration is not?
2. Can the acceleration be constant while the velocity changes sign?
3. If a_x is constant and non-zero, can v_x stay constant?

Decide before any equation

1. Can the velocity be zero while the acceleration is not?
2. Can the acceleration be constant while the velocity changes sign?
3. If a_x is constant and non-zero, can v_x stay constant?

Answers

1. **Yes.** At the top of a thrown ball, $v_y = 0$ but $a_y = -g$.
2. **Yes.** Constant $a_x = -2 \text{ m/s}^2$ takes v_x from $+4$ through 0 to -4 m/s .
3. **No.** $a_x = dv_x/dt$; constant velocity means zero acceleration.

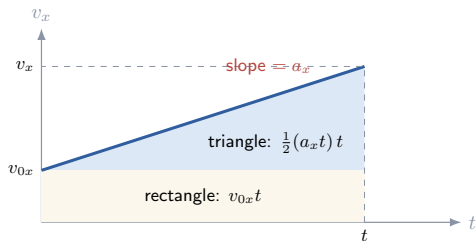
But what if... the acceleration is *constant*, as in these answers? Then the integrals are the same every time, and we can skip the calculus.

When a is constant: skip the calculus

If a_x is constant, $v_x = v_{0x} + a_x t$ is a straight line on a v_x - t graph. The displacement is the area under it:

$$x - x_0 = \underbrace{v_{0x}t}_{\text{rectangle}} + \underbrace{\frac{1}{2}a_x t^2}_{\text{triangle}}$$

Eliminating t between the two results gives the third equation, and the average-velocity form follows from the trapezoid.



But what if... we want the other results too, such as the velocity after a given *distance*? Eliminate t and collect all the equations in one place.

Rule: the constant-acceleration equations

Rule: valid only for constant acceleration along a straight line

- | | | |
|-----|--|--------------|
| (1) | $v_x = v_{0x} + a_x t$ | no $x - x_0$ |
| (2) | $x - x_0 = v_{0x} t + \frac{1}{2} a_x t^2$ | no v_x |
| (3) | $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ | no t |
| (4) | $x - x_0 = \frac{1}{2}(v_{0x} + v_x) t$ | no a_x |
| (5) | $x - x_0 = v_x t - \frac{1}{2} a_x t^2$ | no v_{0x} |

Five variables appear: $x - x_0$, v_{0x} , v_x , a_x and t . Each equation omits exactly one. Derivations are in the supplementary notes.

But what if... we apply these to the previous example, where $a_x = 6.0t - 4.0$ is not constant? Test it with numbers, after the anatomy of two of the equations.

EQUATION – equation (1)

$$v_x = v_{0x} + a_x t$$

Meaning. the velocity changes by the same amount every second.

Variables. v_{0x} initial velocity; a_x constant acceleration; t elapsed time.

Units. $\text{m/s} = \text{m/s} + (\text{m/s}^2)(\text{s})$.

Interpretation. a straight line of slope a_x on the v_x - t graph.

Example. the braking example:
 $t = (0 - 25.0)/(-6.0) = 4.2 \text{ s}$.

EQUATION – equation (3)

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$$

Meaning. links the speeds to the displacement, with no time.

Variables. $x - x_0$ is the displacement; the others as in (1).

Units. $\text{m}^2/\text{s}^2 = \text{m}^2/\text{s}^2 + (\text{m/s}^2)(\text{m})$.

Interpretation. the change in v_x^2 is proportional to the displacement.

Example. the braking example:
 $x - x_0 = 52 \text{ m}$.

Counter-example: the rule has a limit

Apply the constant-acceleration equations to the changing-acceleration example

There $a_x = 6.0t - 4.0$ varies. At $t = 3.0$ s, $a_x = 14$ m/s². Suppose we wrongly used that value as if it were constant:

Wrong (constant $a_x = 14$):

$$v_x = -3.0 + 14(3.0) = +39 \text{ m/s}$$

$$x = 2.0 + (-3.0)(3.0) + \frac{1}{2}(14)(9.0) = 56 \text{ m}$$

Correct (integration):

$$v_x = +12 \text{ m/s}$$

$$x = 2.0 \text{ m}$$

Lesson

Before using equations (1)–(5), ask: *is a_x constant?* If not, integrate.

But what if... a_x *is* constant, and there are five equations to choose from? We need a safe way to pick one: **Toolkit 4**.

TOOLKIT 4 – Choosing a constant-acceleration equation

WHEN TO USE: the acceleration is constant and the motion is along a straight line, with some of $x - x_0$, v_{0x} , v_x , a_x , t known and one or two wanted.

PROCEDURE: **1.** Choose the origin and positive direction (Toolkit 1). **2.** List the known quantities. **3.** Identify the target. **4.** Find the variable that is *neither known nor wanted*. **5.** Choose the equation that omits it. **6.** Solve algebraically first, then substitute with units.

missing	$x - x_0$	v_x	t	a_x	v_{0x}
use	(1)	(2)	(3)	(4)	(5)

WATCH OUT: matching formulas to symbols without checking that a_x is constant, or that the motion is on a line.

CHECK: sign, units and physical meaning; does the answer agree with your prediction and the sketch?

Question

A car moving at 25.0 m/s (90 km/h) brakes with a constant acceleration of magnitude 6.0 m/s^2 until it stops. How far does it travel, and how long does it take?

Predict first

Sketch with $+x$ along the motion. Sign of a_x ? A rough distance if the average speed is about half the initial speed and the stop takes about 4 s ?

Braking distance: solution

Identify: targets $x - x_0$ and t . **Set up:** $+x$ along the motion; $v_{0x} = +25.0 \text{ m/s}$, $v_x = 0$, $a_x = -6.0 \text{ m/s}^2$. Equation (3) omits t ; equation (1) omits $x - x_0$.

Braking distance: solution

Identify: targets $x - x_0$ and t . **Set up:** $+x$ along the motion; $v_{0x} = +25.0 \text{ m/s}$, $v_x = 0$, $a_x = -6.0 \text{ m/s}^2$. Equation (3) omits t ; equation (1) omits $x - x_0$.

Execute: $x - x_0 = \frac{v_x^2 - v_{0x}^2}{2a_x} = \frac{0 - 625}{2(-6.0)} = 52 \text{ m}$ $t = \frac{v_x - v_{0x}}{a_x} = \frac{0 - 25.0}{-6.0} = 4.2 \text{ s}$

Braking distance: solution

Identify: targets $x - x_0$ and t . **Set up:** $+x$ along the motion; $v_{0x} = +25.0 \text{ m/s}$, $v_x = 0$, $a_x = -6.0 \text{ m/s}^2$. Equation (3) omits t ; equation (1) omits $x - x_0$.

Execute: $x - x_0 = \frac{v_x^2 - v_{0x}^2}{2a_x} = \frac{0 - 625}{2(-6.0)} = 52 \text{ m}$ $t = \frac{v_x - v_{0x}}{a_x} = \frac{0 - 25.0}{-6.0} = 4.2 \text{ s}$

Evaluate: average speed $= \frac{1}{2}(25.0 + 0) = 12.5 \text{ m/s}$, and $12.5 \times 4.17 = 52 \text{ m}$, matching equation (4). Units m and s. The estimate ($\approx 50 \text{ m}$) agrees.

Physical meaning

$x - x_0 \propto v_{0x}^2$: doubling the speed quadruples the braking distance.

But what if... the acceleration is the same for every object near the ground, such as a ball thrown upward that rises, stops and falls? One value of a for the whole flight. $\rightarrow$ **free fall**.

Rule: free fall

Near the Earth's surface, neglecting air resistance, all objects have the same acceleration:

$$a_y = -g = -9.80 \text{ m/s}^2 \quad (+y \text{ upward})$$

It holds **throughout** the motion: rising, at the top, and falling.

Predict first

A hammer and a feather are dropped in vacuum. Which lands first?

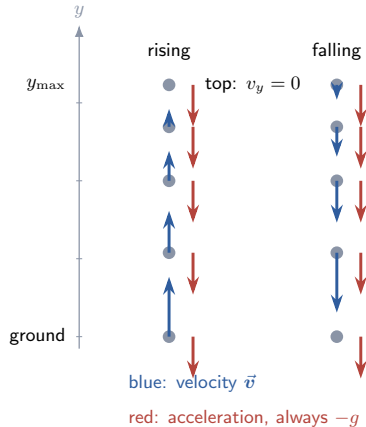
Misconception

“At the top the acceleration is zero.” It is not: $v_y = 0$ momentarily, but $a_y = -g$ still.

Physical meaning

g is the *magnitude*; the sign of a_y comes from the axis you chose (Toolkit 1). With $+y$ downward, $a_y = +g$.

A ball thrown upward: the numbers



A ball is thrown upward at 14.7 m/s from the ground (+y up).

$$\text{Top: } v_y = 0 \Rightarrow t = \frac{14.7}{9.8} = 1.50 \text{ s}$$

$$y_{\text{max}} = \frac{14.7^2}{2(9.8)} = 11.0 \text{ m}$$

Back at ground: $t = 3.00 \text{ s}$ with $v_y = -14.7 \text{ m/s}$.

Neglecting air resistance, and returning to the same height: time up = time down, and the speed is the same at the same height on the way up and down.

But what if... the ball is thrown from the edge of a building, so it lands *below* its launch point? The equation for the landing time is then a quadratic with two roots. Which one is physical?

Question

A ball is thrown **upward** at 14.7 m/s from the edge of a 30.0 m building. How long until it hits the ground, and what is its velocity then? (Neglect air resistance.)

Predict first

Sketch. Is the time more or less than the 3.00 s of a throw from the ground? Sign of v_y at impact?

Two roots: solution

Identify: targets t and v_y . **Set up:** $+y$ up, origin at the ground: $y_0 = 30.0\text{ m}$, $y = 0$, $v_{0y} = +14.7\text{ m/s}$, $a_y = -9.80\text{ m/s}^2$.

Two roots: solution

Identify: targets t and v_y . **Set up:** $+y$ up, origin at the ground: $y_0 = 30.0\text{ m}$, $y = 0$, $v_{0y} = +14.7\text{ m/s}$, $a_y = -9.80\text{ m/s}^2$.

Execute: (2): $0 = 30.0 + 14.7t - 4.9t^2 \Rightarrow 4.9t^2 - 14.7t - 30.0 = 0$

$$t = \frac{14.7 \pm \sqrt{14.7^2 + 4(4.9)(30.0)}}{2(4.9)} = 4.39\text{ s or } -1.39\text{ s}$$

Two roots: solution

Identify: targets t and v_y . **Set up:** $+y$ up, origin at the ground: $y_0 = 30.0\text{ m}$, $y = 0$, $v_{0y} = +14.7\text{ m/s}$, $a_y = -9.80\text{ m/s}^2$.

Execute: (2): $0 = 30.0 + 14.7t - 4.9t^2 \Rightarrow 4.9t^2 - 14.7t - 30.0 = 0$

$$t = \frac{14.7 \pm \sqrt{14.7^2 + 4(4.9)(30.0)}}{2(4.9)} = 4.39\text{ s or } -1.39\text{ s}$$

Only $t = 4.39\text{ s}$ is physical ($t > 0$). Then $v_y = 14.7 - 9.8(4.39) = -28.4\text{ m/s}$.

Evaluate: longer than 3.00 s as predicted; $v_y < 0$ downward; check (3):

$$v_y^2 = 14.7^2 + 2(9.8)(30.0) = 804, \text{ so } |v_y| = 28.4\text{ m/s } \checkmark$$

Interpretation of the negative root

$t = -1.39\text{ s}$ is the time at which the same parabola would have crossed $y = 0$ *before* the throw. Mathematically valid; physically rejected.

Your turn: a stone from a bridge

Independent problem

A stone is dropped from rest from a bridge and hits the water 2.50 s later. How high is the bridge, and how fast is the stone moving when it hits?

Predict first

Sketch with $+y$ up. Sign of v_y at impact? About how far does it fall in the first second, in the second second?

Your turn: a stone from a bridge

Independent problem

A stone is dropped from rest from a bridge and hits the water 2.50 s later. How high is the bridge, and how fast is the stone moving when it hits?

Predict first

Sketch with $+y$ up. Sign of v_y at impact? About how far does it fall in the first second, in the second second?

$$y - y_0 = v_{0y}t + \frac{1}{2}a_y t^2 = 0 + \frac{1}{2}(-9.80)(2.50)^2 = -30.6 \text{ m, so the bridge is 30.6 m high.}$$
$$v_y = 0 + (-9.80)(2.50) = -24.5 \text{ m/s (downward).}$$

Evaluate: negative displacement and velocity, as predicted; check (3):

$$v_y^2 = 2(9.8)(30.6) = 600, |v_y| = 24.5 \text{ m/s } \checkmark$$

Common mistakes and the correct habit

Mistake

Thinking a negative v_x or a negative a_x means slowing down.

Using the constant-acceleration equations when a_x varies.

Setting $a_y = 0$ at the top of a flight.

Keeping the quadratic root with $t < 0$.

Integrating without the initial conditions.

Putting $-g$ into an axis that points down.

Instead

Compare the two signs: the same sign means speeding up.

Ask whether a_x is constant; if not, integrate.

$a_y = -g$ throughout; only v_y is zero there.

Choose the physical root and say why.

Add v_{0x} and x_0 , then check $v(0)$ and $x(0)$.

The sign of a_y comes from the axis you chose.

Connect: Lectures 3 to 5

$a_x = F_x/m$ in Lecture 3; $v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$ becomes the work–energy theorem in Lecture 4; free fall leads to gravitational potential energy in Lecture 5.

Physical meaning

Slopes and areas of graphs, integration with initial conditions, and the constant-acceleration equations with their conditions are the core of one-dimensional kinematics for the rest of the course.

Contact 2: what we can now do

Position

↓ derivative

Velocity

↓ derivative

Acceleration

and back again by integration with initial conditions.

You should be able to

- decide from the signs of v_x and a_x whether the speed increases;
- translate graphs, and integrate or differentiate;
- choose a constant-acceleration equation and say when it applies;
- solve free fall with a consistent sign convention and the physical root.

But what if... the motion is not on a line? A ball kicked at 30° moves both ways at once; a car on a bend has constant speed but $\vec{a} \neq 0$; a walker on a train has two velocities. → **Contact 3.**

Contact 3: Apply

How do the same ideas describe more complex motion?

From motion on a line to motion in a plane and between observers

So far we have modelled motion along a line. We now ask a new question: **what changes when the motion is not along a line, or when the observer is moving?**

From motion on a line to motion in a plane and between observers

So far we have modelled motion along a line. We now ask a new question: **what changes when the motion is not along a line, or when the observer is moving?**

A ball thrown at an angle

projectile motion:
independent components

A car on a bend

circular motion: constant
speed, $\vec{a} \neq 0$

A walker on a train

relative velocity: whose
velocity?

An opening puzzle: the monkey and the dart

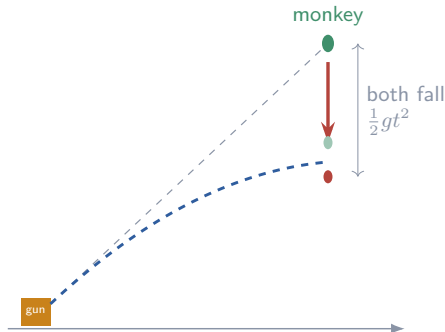
Question

A dart is aimed directly at a monkey hanging in a tree. When the dart leaves the gun, the monkey lets go and falls. Does the dart hit, pass above, or pass below the monkey? (Assume the dart reaches the monkey's line before the monkey reaches the ground.)

Predict first

Commit to an answer. Which of the two experiences gravity?

But what if... the dart moves forward *and* falls at once? We need to know how the horizontal and vertical motions combine.



Rule: independence of components

With uniform gravity and air resistance neglected, the horizontal and vertical component equations can be solved independently. They are linked **only through the common time t** .

Why the dart hits:

Without gravity, both the dart and the monkey would stay on the line of sight. With gravity, both fall the *same* distance $\frac{1}{2}gt^2$ below it in time t . The dart still meets the monkey.

Misconception

No horizontal force acts on a projectile when air resistance is neglected. $a_x = 0$, so v_x is constant.

But what if... we want the height, the time of flight and the range, as numbers? Write both component motions as a **model**.

Projectile motion: the model

Assumptions: neglect air resistance; g constant; flat Earth over the range.

Horizontal ($a_x = 0$):

$$v_x = v_0 \cos \theta$$

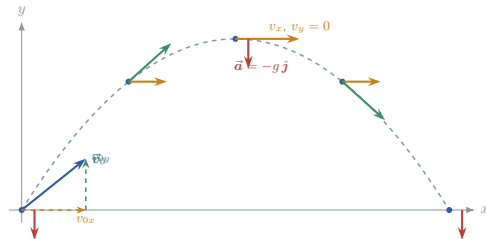
$$x = x_0 + v_0 \cos \theta t$$

Vertical ($a_y = -g$):

$$v_y = v_0 \sin \theta - gt$$

$$y = y_0 + v_0 \sin \theta t - \frac{1}{2}gt^2$$

At the top: $v_y = 0$ but $a_y = -g$.



But what if... we want to know exactly what each symbol, sign and unit means? → **equation anatomy**, then a procedure.

EQUATION – horizontal motion

$$v_x = v_0 \cos \theta$$

$$x = x_0 + v_0 \cos \theta t$$

Meaning. no horizontal acceleration, so the horizontal velocity is constant.

Variables. v_0 launch speed; θ launch angle above the horizontal; t time.

Units. m/s and m.

Interpretation. the range is this constant speed multiplied by the time of flight.

Example. a long hit:

$23.0 \text{ m/s} \times 3.94 \text{ s} = 90.4 \text{ m}$ (unrounded values).

EQUATION – vertical motion

$$v_y = v_0 \sin \theta - gt$$

$$y = y_0 + v_0 \sin \theta t - \frac{1}{2}gt^2$$

Meaning. constant downward acceleration g acting on the vertical motion alone.

Variables. y_0 launch height; $g = 9.80 \text{ m/s}^2$; $+y$ upward.

Units. m/s and m.

Interpretation. a parabola in t ; $v_y = 0$ only at the top.

Example. a long hit: top at
 $t = 19.3/9.80 = 1.97 \text{ s}$.

TOOLKIT 5 – Projectile motion

WHEN TO USE: an object moves freely under gravity alone (air resistance neglected) and its launch velocity is not along the vertical.

PROCEDURE: **1. Sketch** the trajectory and mark the start, top and end. **2. Axes:** $+x$ horizontal, $+y$ up, origin at launch (Toolkit 1). **3. Resolve** $\vec{v}_0$ into v_{0x} and v_{0y} (Lecture 1, Toolkit 3). **4. x -motion:** $a_x = 0$. **5. y -motion:** $a_y = -g$ (Toolkit 4 for the equation). **6. Same time** in both directions; it is the link. **7. Reconstruct** the speed and direction if needed (Lecture 1, Toolkit 4).

WATCH OUT: roots of a quadratic in t : choose the physical one; $v_y = 0$ only at the top, not at landing.

CHECK: does the result agree with the sketch, the prediction and the signs of v_x and v_y ?

But what if... we try it on a real launch from level ground? Predict first, then calculate.

Question

A ball is hit from ground level at 30.0 m/s , 40.0° above the horizontal. Find the time to the highest point, the maximum height, the total time of flight and the horizontal range. (Neglect air resistance; the ground is level.)

Predict first

Sketch. At 45° the range would be $v_0^2/g \approx 92 \text{ m}$; is the range at 40° slightly smaller or larger? What are v_y and a_y at the top?

A long hit: solution

Set up: $v_{0x} = 30.0 \cos 40.0^\circ = 23.0 \text{ m/s}$, $v_{0y} = 30.0 \sin 40.0^\circ = 19.3 \text{ m/s}$.

A long hit: solution

Set up: $v_{0x} = 30.0 \cos 40.0^\circ = 23.0 \text{ m/s}$, $v_{0y} = 30.0 \sin 40.0^\circ = 19.3 \text{ m/s}$.

Execute: top: $v_y = 0 \Rightarrow t = \frac{19.3}{9.80} = 1.97 \text{ s}$; $y_{\max} = \frac{19.3^2}{2(9.80)} = 19.0 \text{ m}$.

Landing: $y = 0 \Rightarrow t = \frac{2(19.3)}{9.80} = 3.94 \text{ s}$ (the other root is $t = 0$).

Range: $x = 23.0 \times 3.94 = 90.4 \text{ m}$ (using unrounded v_{0x} and v_{0y}).

A long hit: solution

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Landing: $y = 0 \Rightarrow t = \frac{2(19.3)}{9.80} = 3.94 \text{ s}$ (the other root is $t = 0$).

Range: $x = 23.0 \times 3.94 = 90.4 \text{ m}$ (using unrounded v_{0x} and v_{0y}).

Evaluate: range slightly less than 92 m, as predicted. At the top $\vec{v} = (23.0 \hat{i} + 0 \hat{j}) \text{ m/s}$ and $a_y = -9.80 \text{ m/s}^2$. Total time is twice the time to the top, as expected for a symmetric path. ✓

But what if... the ball lands *below* its launch height, so the path is no longer symmetric and $2v_{0y}/g$ fails? Use the quadratic and choose the physical root.

Your turn: launched from a height

Independent problem

A ball is kicked at 20.0 m/s , 30.0° above the horizontal, from the edge of a 10.0 m high roof. Find the time to reach the ground, the horizontal distance from the roof, and the velocity (magnitude and direction) at impact.

Predict first

Sketch. Is the time more or less than $2(10.0)/9.80 = 2.04 \text{ s}$? Is the impact velocity above or below the horizontal?

Launched from a height: solution

Set up: $v_{0x} = 17.3 \text{ m/s}$, $v_{0y} = 10.0 \text{ m/s}$, $y_0 = 10.0 \text{ m}$, $y = 0$.

Launched from a height: solution

Set up: $v_{0x} = 17.3 \text{ m/s}$, $v_{0y} = 10.0 \text{ m/s}$, $y_0 = 10.0 \text{ m}$, $y = 0$.

Execute: $0 = 10.0 + 10.0t - 4.9t^2 \Rightarrow t = \frac{10.0 \pm \sqrt{100 + 196}}{9.8} = 2.78 \text{ s}$ or -0.74 s .

Range: $17.3 \times 2.78 = 48.1 \text{ m}$.

$v_y = 10.0 - 9.8(2.78) = -17.2 \text{ m/s}$; $v = \sqrt{17.3^2 + 17.2^2} = 24.4 \text{ m/s}$,

$\theta = \tan^{-1} \frac{17.2}{17.3} = 44.8^\circ$ below the horizontal.

Launched from a height: solution

Set up: $v_{0x} = 17.3 \text{ m/s}$, $v_{0y} = 10.0 \text{ m/s}$, $y_0 = 10.0 \text{ m}$, $y = 0$.

Execute: $0 = 10.0 + 10.0t - 4.9t^2 \Rightarrow t = \frac{10.0 \pm \sqrt{100 + 196}}{9.8} = 2.78 \text{ s}$ or -0.74 s .

Range: $17.3 \times 2.78 = 48.1 \text{ m}$.

$v_y = 10.0 - 9.8(2.78) = -17.2 \text{ m/s}$; $v = \sqrt{17.3^2 + 17.2^2} = 24.4 \text{ m/s}$,

$\theta = \tan^{-1} \frac{17.2}{17.3} = 44.8^\circ$ below the horizontal.

Evaluate: $2.78 \text{ s} > 2.04 \text{ s}$ as predicted; $v_y < 0$; the landing speed exceeds 20.0 m/s because the ball has fallen 10 m below its launch. ✓

But what if... we want the *shape* of the path, not only the landing point? $x(t)$ and $y(t)$ together are the **parametric equations** of the trajectory.

Parametric equations: the trajectory (supporting tool)

$x(t)$ and $y(t)$ together are the **parametric equations** of the path; t is the parameter. Eliminating t gives the shape of the trajectory.

Launch point at $x_0 = y_0 = 0$

$$y = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2$$

a parabola.

Example

$x = 3.0t$, $y = 4.0t - 4.9t^2$ (SI).

Eliminate t and find where $y = 0$ again.

Parametric equations: the trajectory (supporting tool)

$x(t)$ and $y(t)$ together are the **parametric equations** of the path; t is the parameter. Eliminating t gives the shape of the trajectory.

Launch point at $x_0 = y_0 = 0$

$$y = x \tan \theta - \frac{g}{2v_0^2 \cos^2 \theta} x^2$$

a parabola.

Example

$$x = 3.0t, \quad y = 4.0t - 4.9t^2 \text{ (SI).}$$

Eliminate t and find where $y = 0$ again.

$$t = x/3.0, \text{ so } y = 1.33x - 0.544x^2.$$

$$y = 0: x = 1.33/0.544 = 2.45 \text{ m.}$$

But what if... gravity changes both the size and the direction of $\vec{v}$? What if only the *direction* changes, at constant speed? → **circular motion**.

An opening question: acceleration at constant speed?

Poll

A car goes around a circular track at a **constant speed**. Is its acceleration zero?

Predict first

Recall: acceleration is the rate of change of the *velocity vector*. Does the velocity vector stay the same?

An opening question: acceleration at constant speed?

Poll

A car goes around a circular track at a **constant speed**. Is its acceleration zero?

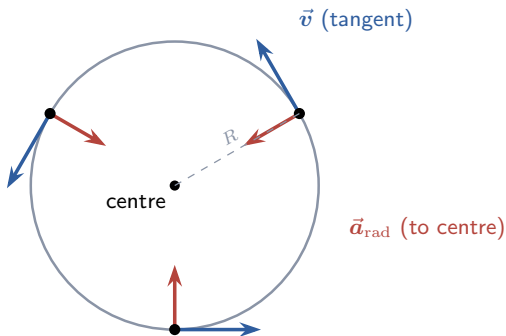
Predict first

Recall: acceleration is the rate of change of the *velocity vector*. Does the velocity vector stay the same?

No. The speed is constant but the direction of $\vec{v}$ changes continuously, so $\vec{a} \neq 0$. This is $a_{\perp}$ from Contact 2, with no $a_{\parallel}$.

But what if... $\vec{a} \neq 0$, but in which direction does it point? Draw $\vec{v}$ at two nearby points.

Uniform circular motion: the geometry



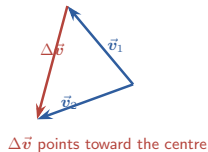
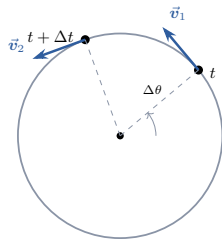
- $\vec{v}$ is tangent to the circle, with constant magnitude v .
- The acceleration has no tangential part: $a_{\text{tan}} = 0$.
- The acceleration points toward the centre: radial, or **centripetal**.

Predict first

If the speed doubles at the same radius, does the acceleration double or quadruple?

But what if... we want its size? Compare $\vec{v}$ at two nearby instants.

Why the acceleration is v^2/R



In a short time Δt the velocity turns through an angle $\Delta\theta$. The arc is $v \Delta t = R \Delta\theta$, and $|\Delta\vec{v}| \approx v \Delta\theta$ points toward the centre. So

$$a_{\text{rad}} = \frac{|\Delta\vec{v}|}{\Delta t} = \frac{v \Delta\theta}{\Delta t} = \frac{v \cdot v \Delta t / R}{\Delta t}$$

Rule: radial acceleration

$$a_{\text{rad}} = \frac{v^2}{R}$$

But what if... we are given the number of turns per second, not v ? Link v to the period and frequency.

Build the relations from one revolution:

angle 2π rad, distance $2\pi R$

- **Period** T : time for one revolution.
- **Frequency** $f = 1/T$ (hertz, $\text{Hz} = \text{s}^{-1}$).
- **Angular speed** $\omega = 2\pi f = 2\pi/T$ (rad/s).
- **Speed**: $v = \frac{2\pi R}{T} = \omega R$.

Rule: uniform circular motion

$$a_{\text{rad}} = \frac{v^2}{R} = \omega^2 R = \frac{4\pi^2 R}{T^2}$$

Physical meaning

Constant speed does not mean zero acceleration. For motion on a circle at constant speed, $\vec{a}$ points to the centre and has magnitude v^2/R .

EQUATION – radial acceleration

$$a_{\text{rad}} = \frac{v^2}{R} = \omega^2 R$$

Meaning. the acceleration that turns the velocity around the circle.

Variables. v speed; R radius; ω angular speed.

Units. m/s^2 ; ω in rad/s .

Interpretation. directed toward the centre; doubling the speed quadruples it.

Example. the bend in the road:
 $20.0^2/80.0 = 5.0 \text{ m/s}^2$.

EQUATION – T , f and ω

$$T = \frac{1}{f}, \quad \omega = 2\pi f = \frac{2\pi}{T}, \quad v = \omega R$$

Meaning. how often, and how fast in angle, the object goes round.

Variables. T period; f frequency; ω angular speed.

Units. T in s; f in $\text{Hz} = \text{s}^{-1}$; ω in rad/s .

Interpretation. one revolution is 2π rad in time T .

Example. the centrifuge: $f = 50 \text{ Hz}$,
 $\omega = 314 \text{ rad/s}$.

TOOLKIT 6 – Circular motion

WHEN TO USE: an object moves on a circle (or a circular arc) and you need its acceleration, speed, period or frequency.

PROCEDURE: **1. Sketch** the circle and mark $\vec{v}$ (tangent) and $\vec{a}$ (to the centre) at one point. **2. Convert** to SI first: $f = \text{rev/min}/60$, $\omega = 2\pi f = 2\pi/T$. **3. Link** $v = \omega R = 2\pi R/T$. **4. Choose** $a_{\text{rad}} = v^2/R$ or $\omega^2 R$, whichever uses the given quantities.

RULE: constant speed does not imply zero acceleration.

WATCH OUT: ω is in rad/s, not rev/min or degrees.

CHECK: $v^2/R = \omega^2 R$ (the two forms agree); $\vec{a}$ points to the centre; its size is reasonable compared with g .

But what if... the car is also speeding up while it rounds the bend? Then $a_{\text{rad}} = v^2/R$ is only *part* of the acceleration.

General curved motion: tangential and normal acceleration (supporting tool)

Rule: general curved motion

$$a_t = \frac{dv}{dt} \quad \text{changes the speed}$$

$$a_n = \frac{v^2}{\rho} \quad \text{changes the direction}$$

ρ is the local radius of curvature of the path.
For uniform circular motion $a_t = 0$ and $\rho = R$,
so $a_n = a_{\text{rad}}$.

Link to Contact 2

Tangential acceleration is the component **parallel** to $\vec{v}$ ($a_{\parallel}$);
normal acceleration is the component **perpendicular** to $\vec{v}$ ($a_{\perp}$).

Predict first

A car speeds up while rounding a bend. Which component is non-zero, a_t , a_n or both?

But what if... we put numbers in? Use the toolkit on a rotating sample, then on a car.

Question

A sample at radius $R = 0.10 \text{ m}$ in a centrifuge rotates at 3000 rev/min . Find the period, the angular speed, the speed of the sample and its radial acceleration in multiples of g .

Predict first

Order of magnitude: much smaller than, similar to, or much larger than g ?

Question

A sample at radius $R = 0.10$ m in a centrifuge rotates at 3000 rev/min. Find the period, the angular speed, the speed of the sample and its radial acceleration in multiples of g .

Predict first

Order of magnitude: much smaller than, similar to, or much larger than g ?

$$f = 3000/60 = 50 \text{ Hz}, \quad T = 1/f = 0.020 \text{ s}, \quad \omega = 2\pi f = 314 \text{ rad/s}, \quad v = \omega R = 31.4 \text{ m/s}.$$
$$a_{\text{rad}} = \omega^2 R = (314)^2(0.10) = 9.9 \times 10^3 \text{ m/s}^2 \approx 1.0 \times 10^3 g.$$

Evaluate: check $v^2/R = 31.4^2/0.10 = 9.9 \times 10^3 \checkmark$; far larger than g , as expected for a centrifuge.

Independent problem

A car travels at a constant 20.0 m/s around a circular bend of radius 80.0 m . Find its radial acceleration, the angular speed, and the time to complete a full circle (if the road were a circle).

Predict first

Draw the direction of $\vec{v}$ and $\vec{a}$ at one point on the circle. Is a larger or smaller than g ?

Your turn: a bend in the road

Independent problem

A car travels at a constant 20.0 m/s around a circular bend of radius 80.0 m . Find its radial acceleration, the angular speed, and the time to complete a full circle (if the road were a circle).

Predict first

Draw the direction of $\vec{v}$ and $\vec{a}$ at one point on the circle. Is a larger or smaller than g ?

$$a_{\text{rad}} = v^2/R = 400/80.0 = 5.0 \text{ m/s}^2 \text{ (toward the centre), about } 0.5 g. \quad \omega = v/R = 0.25 \text{ rad/s.}$$
$$T = 2\pi R/v = 25.1 \text{ s.}$$

Evaluate: check $\omega^2 R = 0.0625 \times 80.0 = 5.0 \text{ m/s}^2 \checkmark$

But what if... a passenger walks along the aisle of the car, or of a train, while it moves? Then we must ask *whose* velocity we mean. $\rightarrow$ **relative velocity**.

An opening situation: whose velocity?

Discussion

A passenger walks along the aisle of a train moving at 20.0 m/s relative to the ground. How fast is the passenger moving?

Predict first

The passenger walks toward the front at 1.5 m/s relative to the train. Faster or slower than the train, relative to the ground? What if the passenger walks toward the rear?

An opening situation: whose velocity?

Discussion

A passenger walks along the aisle of a train moving at 20.0 m/s relative to the ground. How fast is the passenger moving?

Predict first

The passenger walks toward the front at 1.5 m/s relative to the train. Faster or slower than the train, relative to the ground? What if the passenger walks toward the rear?

The same motion has different velocities for different observers.

But what if... the question has no single answer until we say *relative to what?* We need a notation that records whose velocity, and relative to which observer.

Relative velocity: notation and rule

Rule: relative velocity

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$

Read $\vec{v}_{P/A}$ as “the velocity of P relative to A ”.
Also $\vec{v}_{A/B} = -\vec{v}_{B/A}$.

The inner subscript B cancels in the chain: P/B followed by B/A gives P/A .

Predict first

The passenger walks at 1.5 m/s toward the rear of a train moving at 20.0 m/s. Predict $v_{P/G}$ before using the rule.

But what if... we meet this in every problem, with different observers each time? We need a procedure we can repeat: **Toolkit 7**.

TOOLKIT 7 – Relative velocity

WHEN TO USE: a velocity is known relative to one observer and wanted relative to another (train and ground, boat and water, aircraft and air).

PROCEDURE: **1. Identify** the object P and the observers A and B . **2. Write the relation** $\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$ *before* inserting values. **3. Sketch** the vector triangle and choose axes (Toolkit 1). **4. In 2D**, use components (Lecture 1, Toolkits 3 and 5) and reconstruct (Lecture 1, Toolkit 4).

WATCH OUT: do not add speeds: add vectors.

CHECK: the inner subscripts cancel, and the direction agrees with the sketch.

EQUATION – relative velocity

$$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$$

Meaning. the velocity of P seen from A is its velocity seen from B plus the velocity of B seen from A .

Variables. P the moving object; A and B two observers or frames.

Units. m/s for every term.

Interpretation. velocities are vectors, so they add as vectors; the inner subscript B cancels.

Example. the passenger on the train, then the river crossing.

Reading the subscripts

$\vec{v}_{P/T}$: passenger relative to train.

$\vec{v}_{T/G}$: train relative to ground.

$$\vec{v}_{P/G} = \vec{v}_{P/T} + \vec{v}_{T/G}.$$

Back to the train

Passenger P , train T , ground G .

Train relative to ground:

$$v_{T/G,x} = +20.0 \text{ m/s.}$$

Passenger relative to train:

$$v_{P/T,x} = +1.5 \text{ m/s (toward front).}$$

$$v_{P/G,x} = v_{P/T,x} + v_{T/G,x} = 1.5 + 20.0 = 21.5 \text{ m/s.}$$

Toward the rear: $v_{P/T,x} = -1.5 \text{ m/s}$, so $v_{P/G,x} = 18.5 \text{ m/s}$: still moving forward relative to the ground, slower than the train.

Evaluate: the signs follow from the chosen $+x$ (Toolkit 1).

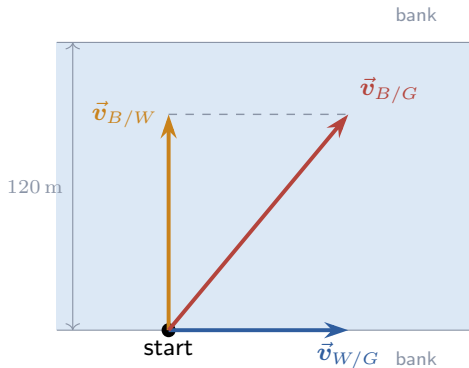
But what if... the velocities are not along one line, such as a boat crossing a flowing river? Use components.

Question

A river flows east at 3.0 m/s relative to the ground and is 120 m wide. A boat has speed 4.0 m/s relative to the water and is pointed due north. Find its velocity relative to the ground, the crossing time and where it lands.

Predict first

Which way is the boat swept? Is the ground speed more or less than 4.0 m/s ?



Crossing a river: solution

Set up: x east, y north. $\vec{v}_{W/G} = 3.0\hat{i}$, $\vec{v}_{B/W} = 4.0\hat{j}$ (m/s). Chain: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$.

Crossing a river: solution

Set up: x east, y north. $\vec{v}_{W/G} = 3.0 \hat{i}$, $\vec{v}_{B/W} = 4.0 \hat{j}$ (m/s). Chain: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$.

Execute: $\vec{v}_{B/G} = (3.0 \hat{i} + 4.0 \hat{j})$ m/s, so $v_{B/G} = 5.0$ m/s at $\tan^{-1}(3.0/4.0) = 36.9^\circ$ east of north.

Crossing time depends only on the northward motion: $t = \frac{120}{4.0} = 30$ s. Downstream drift:
 $3.0 \times 30 = 90$ m.

Set up: x east, y north. $\vec{v}_{W/G} = 3.0 \hat{i}$, $\vec{v}_{B/W} = 4.0 \hat{j}$ (m/s). Chain: $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$.

Execute: $\vec{v}_{B/G} = (3.0 \hat{i} + 4.0 \hat{j})$ m/s, so $v_{B/G} = 5.0$ m/s at $\tan^{-1}(3.0/4.0) = 36.9^\circ$ east of north.

Crossing time depends only on the northward motion: $t = \frac{120}{4.0} = 30$ s. Downstream drift: $3.0 \times 30 = 90$ m.

Evaluate: swept east, ground speed $5.0 > 4.0$ m/s, as predicted; the current adds sideways motion but does not change the crossing time. $\sqrt{90^2 + 120^2} = 150$ m = 5.0×30 ✓

But what if... the boat must land at the point *directly opposite*? Pointing north would be swept away. Which way must we point?

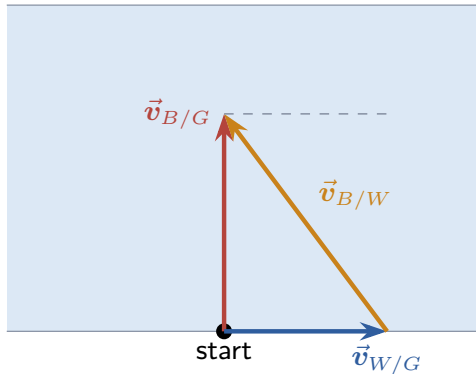
Your turn: heading straight across

Independent problem

The same river (3.0 m/s east, 120 m wide). A boat with speed 5.0 m/s relative to the water must reach the point directly opposite its start. In which direction must it be pointed, and how long does the crossing take?

Predict first

Point the boat upstream or downstream? Is the ground speed more or less than 5.0 m/s?



Heading straight across: solution

Set up: we need $\vec{v}_{B/G}$ due north: $v_{B/G,x} = 0$. From $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$, with $\vec{v}_{B/W}$ at angle φ west of north:

Heading straight across: solution

Set up: we need $\vec{v}_{B/G}$ due north: $v_{B/G,x} = 0$. From $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$, with $\vec{v}_{B/W}$ at angle φ west of north:

Execute: x : $-5.0 \sin \varphi + 3.0 = 0 \Rightarrow \sin \varphi = 0.60$, $\varphi = 36.9^\circ$ **west of north** (upstream).

y : $v_{B/G,y} = 5.0 \cos 36.9^\circ = 4.0 \text{ m/s}$; time $t = 120/4.0 = 30 \text{ s}$.

Heading straight across: solution

Set up: we need $\vec{v}_{B/G}$ due north: $v_{B/G,x} = 0$. From $\vec{v}_{B/G} = \vec{v}_{B/W} + \vec{v}_{W/G}$, with $\vec{v}_{B/W}$ at angle φ west of north:

Execute: x : $-5.0 \sin \varphi + 3.0 = 0 \Rightarrow \sin \varphi = 0.60$, $\varphi = 36.9^\circ$ **west of north** (upstream).

y : $v_{B/G,y} = 5.0 \cos 36.9^\circ = 4.0 \text{ m/s}$; time $t = 120/4.0 = 30 \text{ s}$.

Evaluate: upstream, as predicted; the ground speed $4.0 \text{ m/s} < 5.0 \text{ m/s}$ because part of the boat's speed cancels the current. The same triangle as the previous example. ✓

More crosswind problems (cf. Young and Freedman, Examples 3.14 and 3.15) are on the tutorial sheet.

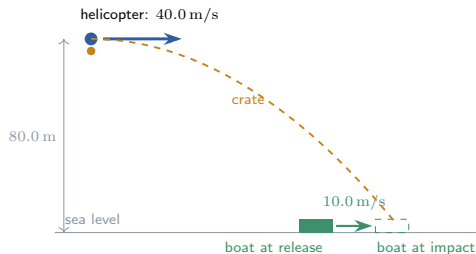
But what if... a problem uses several of these ideas at once: a falling object seen from a moving boat, or a circular path that ends in a projectile? Which model, and when? → **integrated applications**.

Integrated problem: dropping a crate

Application problem: decide what physics applies

A helicopter flies horizontally at 40.0 m/s , 80.0 m above the sea. A boat below moves in the same direction at 10.0 m/s . A crate released from the helicopter is to land on the boat.

- (a) How far ahead of the crate (horizontally) must the boat be at release?
- (b) With what speed does the crate hit the boat, as seen from the boat?



Predict first

IDENTIFY $\rightarrow$ SET UP (frame? axes? sketch) $\rightarrow$ PREDICT $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE.

Identify: the crate is a projectile (free fall with initial horizontal velocity); the boat is another frame.

Set up: $+x$ forward, $+y$ up; crate $\vec{v}_{C/G} = (40.0, 0)$ at release. Predict: the crate gains horizontally on the boat for the whole fall time.

Identify: the crate is a projectile (free fall with initial horizontal velocity); the boat is another frame.

Set up: $+x$ forward, $+y$ up; crate $\vec{v}_{C/G} = (40.0, 0)$ at release. Predict: the crate gains horizontally on the boat for the whole fall time.

Execute (a): fall time from $y = 80.0$ m: $t = \sqrt{2(80.0)/9.80} = 4.04$ s (Toolkit 5).
Relative horizontal speed $v_{C/B,x} = 40.0 - 10.0 = 30.0$ m/s (Toolkit 7), so the boat must be $30.0 \times 4.04 = 121$ m ahead.

Identify: the crate is a projectile (free fall with initial horizontal velocity); the boat is another frame.

Set up: $+x$ forward, $+y$ up; crate $\vec{v}_{C/G} = (40.0, 0)$ at release. Predict: the crate gains horizontally on the boat for the whole fall time.

Execute (a): fall time from $y = 80.0$ m: $t = \sqrt{2(80.0)/9.80} = 4.04$ s (Toolkit 5).
Relative horizontal speed $v_{C/B,x} = 40.0 - 10.0 = 30.0$ m/s (Toolkit 7), so the boat must be $30.0 \times 4.04 = 121$ m ahead.

(b): $v_{C/B,y} = -9.80(4.04) = -39.6$ m/s, so $v_{C/B} = \sqrt{30.0^2 + 39.6^2} = 49.7$ m/s.

Evaluate: units m and m/s; the crate relative to the boat has the same fall time but a smaller horizontal speed; 4.04 s is plausible for an 80 m fall ($\sqrt{2h/g}$). ✓

Decide what physics applies; the method is not given

A stone on a string 0.60 m long is whirled at 2.5 revolutions per second in a horizontal circle, 1.40 m above level ground. The string breaks while the stone is moving horizontally.

- (a) What is the stone's radial acceleration just before the string breaks?
- (b) How far does the stone travel horizontally before it lands, and what is its speed on landing? (Neglect air resistance.)

Predict first

IDENTIFY → SET UP → PREDICT → EXECUTE → EVALUATE. Which model describes each stage? Is the radial acceleration much larger or much smaller than g ?

Your turn: an integrated problem, solution

Identify: stage 1 is uniform circular motion (Toolkit 6); stage 2 is a projectile launched horizontally from 1.40 m (Toolkit 5).

Set up: $f = 2.5 \text{ Hz}$, $R = 0.60 \text{ m}$. At the break, $\vec{v}$ is tangent and horizontal, so $v_{0x} = v$ and $v_{0y} = 0$.

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Execute (a): $\omega = 2\pi f = 15.7 \text{ rad/s}$, $v = \omega R = 9.42 \text{ m/s}$,

$$a_{\text{rad}} = \frac{v^2}{R} = \frac{88.8}{0.60} = 148 \text{ m/s}^2 \approx 15 g.$$

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(b): fall time $t = \sqrt{2(1.40)/9.80} = 0.535 \text{ s}$; horizontal distance $9.42 \times 0.535 = 5.04 \text{ m}$.
 $v_y = -9.80(0.535) = -5.24 \text{ m/s}$, so $v = \sqrt{9.42^2 + 5.24^2} = 10.8 \text{ m/s}$, at
 $\tan^{-1}(5.24/9.42) = 29.1^\circ$ below the horizontal.

Evaluate: much larger than g , as predicted; the landing speed exceeds the break-off speed because the stone has fallen; check $\sqrt{2gh + v^2} = \sqrt{27.4 + 88.8} = 10.8 \text{ m/s}$ ✓

Common mistakes and the correct habit

Mistake

A horizontal force on a projectile.

Using different times for x and y .

Setting $a = 0$ at constant speed on a circle.

Adding speeds in relative motion.

Using rev/min or degrees in $v = \omega R$.

Setting $v_y = 0$ at landing.

Instead

None is needed: $a_x = 0$ and v_x is constant (no air resistance).

The common time is the only link between the components.

$a = v^2/R$ toward the centre.

Add velocities as vectors with the subscript chain.

Convert to rad/s first.

$v_y = 0$ only at the top of the flight.

Projectiles

Lecture 5 finds the speed at any height by energy conservation, with no time; Lecture 7 treats collisions in two dimensions with the same component method.

Circular motion

Lecture 3 supplies the force behind $a_{\text{rad}} = v^2/R$ (centripetal force); Lecture 8 extends ω , T and f to rotating bodies.

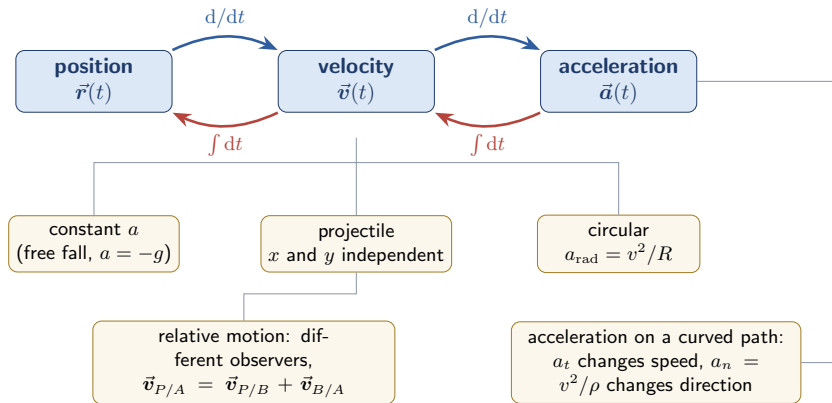
Relative velocity

Lecture 7: a collision looks different in different frames, and the velocity of the centre of mass is the natural frame in which to describe it.

Physical meaning

Independent components, the constant-acceleration equations and $\vec{v} = d\vec{r}/dt$ remain the core of the motion toolbox for the rest of the course.

Week 2 concept map



Key ideas

- **Position** is a vector from the origin you choose; displacement $\Delta \vec{r} = \vec{r}_2 - \vec{r}_1$ does not depend on the origin; distance travelled is a non-negative scalar.
- **Velocity** $\vec{v} = d\vec{r}/dt$ is a vector tangent to the path; speed $v = |\vec{v}|$ is its magnitude; v_x is a signed number.
- **Acceleration** $\vec{a} = d\vec{v}/dt$: the part parallel to $\vec{v}$ changes the speed; the perpendicular part changes the direction.
- Slope and area link the $x-t$, $v-t$, $a-t$ graphs. Along a line, v_x and a_x with the same sign mean speeding up.
- **The constant-acceleration equations hold only when a_x is constant along a line; otherwise integrate, with initial conditions.**
- Free fall without air resistance: $a_y = -g$ throughout the flight, including the top.
- **In projectile motion (no air resistance) the horizontal and vertical motions are independent and linked only through the time.**
- Constant speed on a circle still means $a = v^2/R \neq 0$, directed to the centre.
- Velocities add as vectors, with the subscript chain $\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$.
- Habit: sketch, predict, calculate, evaluate.

Reference tools: equations, procedures and toolkits follow. Definitions: vocabulary and anatomy slides. Mistakes: last slide of each contact.

What you should remember: the chain, Contact 1 (part 1)

Concept	Limitation that motivates the next	Leads to
Opening journey	"how far?" has two answers: 60 m from the start, 140 m of road	Vocabulary
Vocabulary	the same word names a vector in one place and a number in another	Notation
Notation	the sign of v_x depends on which way is positive	Position, displacement
Position, displacement	$\Delta x = +60$ m but distance 140 m; reversing the axis flips every sign	Toolkit 1
Toolkit 1	how rapidly did the position change over 30 s?	Average velocity
Average velocity, dragster	+2.0 m/s hides +5.0 then -4.0 m/s; the dragster has no single speed	Instantaneous velocity

What you should remember: the chain, Contact 1 (part 2)

Concept	Limitation that motivates the next	Leads to
Instantaneous velocity	the chord table only approaches 12 m/s; we need it exactly	Derivative, anatomy
Velocity anatomy, concept check	the motion is not along one line	Vectors
Vectors	$x(t)$, $y(t)$, $z(t)$ given as formulas: how do we get $\vec{v}$?	Toolkit 2
Toolkit 2, helix example	only a graph, no formula	Toolkit 3
Toolkit 3	velocity itself changes: +5.0 to -4.0 m/s in 2.0 s	Acceleration (Contact 2)

What you should remember: the chain, Contact 2 (part 1)

Concept	Limitation that motivates the next	Leads to
Opening question	we need the rate of change of velocity itself	Average acceleration
Average acceleration	the runner's -4.5 m/s^2 is an average over 2.0 s	Instantaneous acceleration
Instantaneous acceleration	$a_x < 0$ throughout the turn, yet the runner first slows, then speeds up	Sign rule
Sign rule	can we see it with no numbers?	Motion diagrams
Motion diagram	the velocity changes direction as well as size	Acceleration along, across
Acceleration along, across	a $v-t$ graph of straight segments: slopes give a_x , but displacement?	Toolkit 3 completed
Toolkit 3 completed	the graph is a smooth curve given by a formula	Integration

What you should remember: the chain, Contact 2 (part 2)

Concept	Limitation that motivates the next	Leads to
Integration, Toolkit 2 completed	integrating every time is slow when a_x is constant	Constant-acceleration equations
Constant-acceleration equations	they fail if we use them with $a_x = 6.0t - 4.0$: 39 against 12 m/s	Counter-example
Counter-example	with a_x constant, five equations: which one?	Toolkit 4
Toolkit 4, braking distance	every object near the ground has the same a_y	Free fall
Free fall, ball thrown upward	landing below the launch point gives two roots	Two roots
Two roots, stone	the motion is not on one line	Contact 3

What you should remember: the chain, Contact 3 (part 1)

Concept	Limitation that motivates the next	Leads to
Monkey and dart	how do horizontal and vertical motions combine?	Independent components
Independent components	height, time of flight and range as numbers	Projectile model, Toolkit 5
Model, anatomy, Toolkit 5	a real launch from level ground	Long hit
Long hit	landing below the launch height: not symmetric	Launched from a height
Launched from a height	the shape of the whole path	Parametric equations
Parametric equations	only the direction of $\vec{v}$ changes, at constant speed	Circular motion
Circular geometry	how large is $\vec{a}$?	$a_{\text{rad}} = v^2/R$

What you should remember: the chain, Contact 3 (part 2)

Concept	Limitation that motivates the next	Leads to
$a_{\text{rad}} = v^2/R$	the data are revolutions per second, not v	Period, frequency, ω
Period, frequency, Toolkit 6	the car also speeds up on the bend	Tangential and normal acceleration
Tangential, normal	put numbers in	Centrifuge, bend
Bend in the road	a passenger walks inside the moving car: whose velocity?	Relative velocity
Relative velocity, Toolkit 7	the velocities are not along one line	River crossing
River crossing	several ideas at once: which model, and when?	Integrated applications

Reference tools: core equations

Idea	Equation	Conditions and units
Displacement	$\Delta x = x_2 - x_1, \quad \Delta \vec{r} = \vec{r}_2 - \vec{r}_1$	m; final minus initial
Velocity	$\vec{v} = d\vec{r}/dt, \quad v_{av,x} = \Delta x / \Delta t$	m/s; slope of $x-t$
Acceleration	$\vec{a} = d\vec{v}/dt = d^2\vec{r}/dt^2$	m/s ² ; slope of v_x-t
Speeding up	v_x and a_x same sign: speeds up; opposite: slows	motion along a line
Integration	$v_x = v_{0x} + \int a_x dt, \quad x = x_0 + \int v_x dt$	needs initial conditions
Constant a_x	$v_x = v_{0x} + a_x t, \quad v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	only if a_x is constant, straight line
Free fall	$a_y = -g, \quad g = 9.80 \text{ m/s}^2$	+y upward; no air resistance
Projectile	$a_x = 0, \quad a_y = -g; \text{ same } t$	$v_{0x} = v_0 \cos \theta, v_{0y} = v_0 \sin \theta$
Circular motion	$a_{rad} = v^2/R = \omega^2 R, \quad v = \omega R$	toward the centre; $\omega = 2\pi f = 2\pi/T$
Curved path	$a_t = dv/dt, \quad a_n = v^2/\rho$	speed and direction changes
Relative velocity	$\vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$	vector sum; m/s

MASTER PROCEDURE – MOTION

1. **IDENTIFY** Which object, which interval, what is asked?
2. **FRAME** Relative to which observer is the motion described?
3. **CHOOSE AXES** Origin and positive directions (Toolkit 1).
4. **SKETCH** Trajectory or motion diagram; add $\vec{v}$ and $\vec{a}$.
5. **PREDICT** Signs, direction, trend, rough size, physical root.
6. **SELECT MODEL** General $\vec{r}, \vec{v}, \vec{a}$? Constant a ? Free fall? Projectile? Circular? Relative motion?
7. **EXECUTE** Apply the matching toolkit; keep signs and units.
8. **EVALUATE** Units, sign, direction, graph or sketch, assumptions, reasonableness.

If the result disagrees with the sketch or prediction, investigate before accepting the answer.

Inside the course framework

IDENTIFY (steps 1–2):
what is asked?

SET UP (steps 3–6): axes,
sketch, predict, model.

EXECUTE (step 7): apply
the matching toolkit.

EVALUATE (step 8):
check the result.

Physical meaning

Step 6 tells you which of the
seven toolkits to open.



THE SEVEN TOOLKITS

- | | | |
|---|--|--|
| 1 | Signs and coordinates | origin and positive direction first; signs relative to your axis; final minus initial |
| 2 | Calculus of motion | differentiate $\vec{r} \rightarrow \vec{v} \rightarrow \vec{a}$; integrate back with initial conditions |
| 3 | Reading motion graphs | slope and area for $x-t$, $v-t$, $a-t$; translate both ways |
| 4 | Choosing a constant-a equation | list known, wanted, missing; confirm a constant |
| 5 | Projectile motion | resolve; independent x and y ; same time; physical root |
| 6 | Circular motion | $\vec{v}$ tangent, $\vec{a}$ to the centre; $a_{\text{rad}} = v^2/R = \omega^2 R$ |
| 7 | Relative velocity | subscript chain; vectors, not speeds |

The master procedure sits above these. Lecture 1 toolkits (components, reconstructing a vector, adding by components) are reused, not renamed.

Checklist: can I ...

- ☐ tell displacement from distance, and velocity from speed, in a given journey?
- ☐ choose an origin and positive direction, and give each sign its meaning?
- ☐ find $\vec{v}$ and $\vec{a}$ from $\vec{r}(t)$ by differentiating each component?
- ☐ find $v_x(t)$ and $x(t)$ from $a_x(t)$ by integrating, with initial conditions?
- ☐ read slopes and areas on $x-t$, $v-t$ and $a-t$ graphs, and sketch one from a description?
- ☐ decide from the signs of v_x and a_x whether an object speeds up or slows down?
- ☐ choose a constant-acceleration equation, and show that a_x is constant first?
- ☐ solve a free-fall problem with a consistent sign convention, and reject the non-physical root?
- ☐ solve a projectile problem with x and y linked only by time?
- ☐ explain why uniform circular motion has acceleration, and calculate a_{rad} ?
- ☐ find a velocity relative to a second observer with the subscript chain?
- ☐ use IDENTIFY $\rightarrow$ SET UP $\rightarrow$ EXECUTE $\rightarrow$ EVALUATE, and choose the model in an unfamiliar problem?

Next

Tutorial sheet: the train and police-car problems and extra graph, free-fall, projectile, circular and relative-velocity problems. Supplementary notes: derivations, the calculus refresher and the parabola extension.